E-commerce System with Mock Payment Integration: Project Plan

1.Introduction

The advancement of digital technologies has significantly transformed the retail industry, with e-commerce platforms becoming a central component of modern business operations. This project focuses on the design and development of a web-based e-commerce system that incorporates a mock payment integration to simulate real-world online transactions. The system will allow users to browse products, manage a shopping cart, enter shipping details, and complete a checkout process using a simulated payment gateway. The purpose of this project is to apply software engineering principles in building a functional, scalable, and user-friendly system while gaining practical experience in full-stack development.

The background of this project lies in the increasing demand for digital commerce solutions and the need for students to develop practical systems that mirror real-world applications. The project aims to demonstrate competencies in system design, database management, API integration, and user experience design. The objectives include developing a complete e-commerce workflow, implementing a mock payment system, and ensuring the system meets both functional and non-functional requirements.

2. User/Client Involvement

User or client involvement is essential to ensure that the system meets its intended purpose. In this project, users—primarily online shoppers—will be involved through usability testing and feedback sessions. They will evaluate the interface design, ease of navigation, and the effectiveness of the checkout process. Feedback collected will be used to refine system features and improve overall user experience. The client, in this case represented by the project requirements, will provide the necessary guidelines, resources, and expectations for the system.

3. Risks

Despite careful planning, the project is subject to several risks. These include potential challenges in integrating the mock payment system, poor database structure leading to inefficiencies, and possible time constraints due to the limited project duration of eight weeks. Additionally, there is a risk of developing a system that does not meet user expectations in terms of usability. To address these risks, mitigation strategies such as incremental development, continuous testing, proper documentation, and effective time management will be implemented.

4. Standards, Guidelines and Procedures

The project will adhere to established standards, guidelines, and procedures to ensure quality and consistency. Coding standards will emphasize readability, modularity, and maintainability. Version control systems will be used to track changes and manage development progress.

Documentation will be maintained throughout the project to ensure clarity and ease of understanding. Furthermore, basic security practices, including input validation and safe handling of user data, will be followed to enhance system reliability.

5. Organization of the Project

In terms of organization, the project typically involves roles such as project manager, developer, tester, and designer. However, since this is an individual project, all roles will be performed by a single developer. This requires effective planning and discipline to ensure that all aspects of the project are completed successfully within the given timeframe.

6. Project Phases

The project will follow an iterative development model inspired by Agile methodologies, allowing for flexibility and continuous improvement. The project is scheduled to be completed over a period of eight weeks, with each week focusing on specific deliverables. In Week 1, project planning and requirement analysis will be conducted. Week 2 will focus on system design, including architecture and database schema. Weeks 3 and 4 will involve frontend development, where the user interface and user experience components are built. Weeks 5 and 6 will focus on backend development and integration, including API development and database connectivity. Week 7 will be dedicated to testing, debugging, and refinement of the system. Finally, Week 8 will involve deployment, final evaluation, and documentation. Key milestones will include the completion of the user interface, functional backend, successful payment simulation, and deployment of the system.

7. Requirements Analysis and Design

Requirements analysis and design are critical to the success of the project. Functional requirements include the ability for users to browse products, add and remove items from a shopping cart, enter shipping details, complete a checkout process, and receive an order confirmation. Non-functional requirements include system performance, security, reliability, and ease of use. The system architecture will consist of a frontend interface, a backend server for processing logic, and a database for storing information such as products, users, and orders. This structured design ensures efficient communication between system components.

8. Implementation

The implementation phase will involve the use of modern web development technologies. The front end will be developed to provide an interactive and responsive user interface, while the backend will handle business logic and communication with the database. A mock payment system will be integrated to simulate real transactions without involving actual financial processing. Development will follow a step-by-step approach, beginning with the user interface, followed by backend logic, and finally system integration.

9. Testing

Testing is an essential component of the project to ensure functionality and reliability. Different testing methods will be employed, including unit testing to verify individual components, integration testing to ensure that system components work together, and user testing to evaluate usability. Test scenarios will include adding items to the cart, completing a transaction, handling failed payments, and verifying order confirmations. Regular testing will help identify and resolve issues early in the development process.

10. Resources

The project requires both hardware and software resources. Hardware requirements include a computer and reliable internet access. Software tools include a code editor, a web browser for testing, and a version control system for managing code. These resources are essential for supporting the development and deployment of the system.

11. Quality Assurance

Quality assurance will be maintained throughout the project lifecycle to ensure that the final product meets the required standards. This will involve continuous testing, code reviews, and validation against the initial requirements. Any defects identified will be corrected promptly to maintain system integrity.

12. Changes

Changes during the development process are inevitable and must be managed effectively. A structured change management process will be followed, which includes identifying the need for change, analyzing its impact on the system, implementing the necessary modifications, and testing the updated system. Proper documentation of all changes will be maintained to ensure transparency and accountability.

In conclusion, this project provides a valuable opportunity to apply software engineering principles in a practical context. By developing an e-commerce system with mock payment integration, the project demonstrates the ability to design, implement, and evaluate a real-world application. With careful planning, adherence to standards, and effective time management over the eight-week period, the project is expected to deliver a functional and high-quality system that meets both academic and practical objectives.

Below is a Gantt chart that shows the project schedule over the 8 weeks we intend on finishing the project:

Documentation								
Debugging								
Integration								
Development								
Development								
System Design								
Requirement								
	Week 1-2	Week 2-3	Week 3-4	Week 4-5	Week 5-6	Week 6-7	Week 7-8	Week 8-9

Followed by the Gantt chart is a PERT diagram that visually maps the sequence, timing, and dependencies of tasks in the project: